THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — SCIENCE · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. To find whether the volume of water affects its boiling time, a student boils 100 mL in a steel pot on a strong flame and 200 mL in a glass beaker on a weak flame. Why is this test unfair? 1

(a) the water is heated at all
(b) the volumes of water are different
(c) time is measured with a clock
(d) the container and the flame also differ along with the volume

2. A scientist finds a plant with parallel veins, fibrous roots and a soft stem. Which is most probable? 1

(a) a peepal tree **(b)** a grass **(c)** a rose bush **(d)** a mango tree

3. A food gives a blue-black colour with iodine and an oily patch on paper. It contains: 1

(a) fat only **(b)** starch only
(c) starch and fat **(d)** protein and fibre

4. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) all are iron
(b) all are plastic
(c) all are magnets
(d) P and R are magnets, Q is magnetic material

5. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 320 cm **(b)** 100 cm **(c)** 160 cm **(d)** 200 cm

6. A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

(a) R **(b)** S **(c)** P **(d)** Q

7. Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

(a) 48 °C **(b)** 52 °C **(c)** 50 °C **(d)** 54 °C

8. When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

(a) the thermometer is faulty **(b)** ice cannot be heated
(c) the heater is weak **(d)** heat is used to change the state

9. A mixture of iron filings, common salt, and fine sand is given. Which sequence is the most efficient? 1

(a) add water, evaporate, magnet
(b) filter, magnet, sieve
(c) magnet, add water and filter, evaporate filtrate
(d) evaporate first, filter, magnet

10. A candle burns, melts, gives out waste gases and its flame responds to wind. Why is it still non-living? 1

(a) it does not grow or reproduce on its own **(b)** it melts
(c) it is hot **(d)** it makes light

11. A student collected 10 mL of gas from a jar over water after a candle burnt out. The candle extinguished because: 1

(a) the jar broke **(b)** all nitrogen was used
(c) no gas remained **(d)** most oxygen was used

12. If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

(a) 8 light-years **(b)** 8 light-hours **(c)** 8 light-seconds **(d)** 8 light-minutes

13. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** sunburn causes ice cream sales
(c) it is magic **(d)** ice cream causes sunburn

14. Which of the following is a correct inference about a tiger's eyes in front of the face? 1

(a) helps fly **(b)** helps see all sides
(c) helps judge distance for hunting **(d)** helps to swim

15. Why do nutritionists advise limiting packaged drinks? 1

(a) they contain much sugar and few nutrients
(b) they have vitamins
(c) they have fibre
(d) they contain protein

16. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not **(b)** both are shiny
(c) both are metals **(d)** both rust

17. In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

(a) time period is 1.8 s in both **(b)** he is wrong
(c) time period differs **(d)** 36 s vs 18 s differ

18. A boy tests four materials A, B, C, D with a torch. Light passes fully through A, partly through B, not at all through C and D. Which can be used for a window to see out and still keep privacy? 1

(a) C **(b)** A **(c)** B **(d)** D

19. Rod A conducts heat twice as fast as rod B. If wax pins fall from A in 20 s from the heated end, the same pin on B would fall in about: 1

(a) 40 s **(b)** 20 s **(c)** 5 s **(d)** 10 s

20. Water of 100 g at 0 °C is converted to ice at 0 °C. The mass of ice formed is: 1

(a) more than 100 g **(b)** 100 g **(c)** 0 **(d)** less than 100 g

21. 100 g of a mixture contains 20 g of iron, 30 g of sand and 50 g of salt. After separation, the mass of sand obtained is expected to be: 1

(a) 50 g **(b)** 100 g **(c)** 20 g **(d)** 30 g

22. Out of a tree, a bird and a pebble, which one is non-living? 1

(a) pebble **(b)** bird
(c) tree **(d)** both bird and tree

23. Which of these actions improves soil fertility naturally? 1

(a) adding plastic **(b)** adding compost **(c)** removing worms **(d)** burning plastics

24. Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

(a) 360 kg **(b)** 6 kg **(c)** 10 kg **(d)** 60 kg

25. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) toss a coin **(b)** compare the methods and repeat carefully

(c) ignore both **(d)** believe one group

26. In a forest, if all insects vanish, what is most likely to happen? 1

(a) animals will grow bigger **(b)** forest will grow faster

(c) many plants will fail to pollinate **(d)** all plants will grow faster

27. A paper rubbed with a food sample shows a translucent oily patch that stays even after drying. The food contains: 1

(a) fat **(b)** starch **(c)** fibre **(d)** sugar

28. To make a magnet from an iron bar by the single-touch method, the bar should be stroked: 1

(a) by hitting it with the magnet

(b) back and forth with both poles

(c) in a circle with the magnet

(d) always in the same direction with the same pole of a magnet

29. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 12.5 cm **(b)** 14.5 cm **(c)** 2.0 cm **(d)** 10.5 cm

30. Why does a heavy ship made of steel float? 1

(a) steel is lighter than water **(b)** ships have no weight

(c) its shape displaces much water **(d)** water is sticky

31. The best conductor-insulator pair for a cooking pot and its handle is: 1

(a) glass and iron **(b)** plastic and aluminium

(c) aluminium and plastic **(d)** wood and plastic

32. A student claims that the water in a cup disappeared. The correct statement is: 1

(a) it changed to vapour and mixed with air **(b)** it became oxygen

(c) it vanished **(d)** it was destroyed

33. Why is alum added during water purification before filtration? 1

(a) it kills all germs **(b)** it adds taste

(c) it dissolves salt **(d)** it makes tiny particles clump and settle

34. A dry seed lies unchanged in a packet for months but sprouts after being watered. This shows that: 1

(a) water makes seeds alive from stone

(b) a living thing can stay inactive until conditions suit it

(c) seeds are non-living

(d) sprouting is random

35. A town has 1,00,000 vehicles each emitting smoke. Which step reduces the problem most? 1

(a) more public transport **(b)** less trees

(c) more cars **(d)** more roads only

36. Which of the following explains day and night on Earth? 1

(a) Earth rotates on its axis **(b)** the Sun moves around the Earth

(c) Earth revolves around the Sun once a day **(d)** the Moon blocks the Sun daily

37. In an experiment on how salt affects the boiling point, two things were changed: salt amount and heater power. The error is: 1

(a) more than one variable was changed **(b)** too many readings

(c) salt is a solid **(d)** water is cold

38. An animal with webbed feet, lungs and moist skin that lives near ponds and jumps on land is most likely a: 1

(a) lion **(b)** crow **(c)** frog **(d)** camel

39. A student tested three foods P, Q, R with iodine. Only Q turned blue-black. Which statement is correct? 1

(a) All three contain starch **(b)** P and R contain starch
(c) Q contains fat **(d)** Q contains starch

40. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) south-seeking pole **(b)** north-seeking pole
(c) neutral centre **(d)** non-magnetic part

41. A road is 2.7 km. A runner has run 1850 m. The remaining distance is: 1

(a) 650 m **(b)** 1850 m **(c)** 750 m **(d)** 850 m

42. In the experiment above, the worst conductor is: 1

(a) R **(b)** S **(c)** P **(d)** Q

43. Hot water at 80 °C is poured into three cups: metal, ceramic and plastic. After 5 minutes the water is coolest in the: 1

(a) metal cup **(b)** plastic cup **(c)** ceramic cup **(d)** all equal

44. A student heats 100 g of ice and records temperatures. The graph has a flat portion at 0 °C and another at 100 °C. The flat portions represent: 1

(a) cooling **(b)** melting and boiling
(c) freezing **(d)** heating of solid and liquid

45. In the mixture above, after the first step the residue on the filter paper is: 1

(a) salt **(b)** water **(c)** iron **(d)** sand

46. A student observes a crystal growing in a saturated solution. Which statement is true? 1

(a) the crystal grows by outside deposition, not from within
(b) it respires
(c) it is living
(d) it reproduces

47. In that experiment, water rises in the jar by about one fifth because: 1

(a) nitrogen forms one-fifth **(b)** water is heavy
(c) oxygen forms about one-fifth of air **(d)** jar is wide

48. A student draws the Moon at the same time each night and sees the bright part grows. The Moon is: 1

(a) new **(b)** waning **(c)** missing **(d)** waxing

49. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) amount of light **(b)** type of plant **(c)** amount of water **(d)** amount of soil

50. Why do prey animals like deer have eyes on the sides of the head? 1

(a) to swim **(b)** to see only forward
(c) to dig **(d)** wide field of view to spot predators

END OF QUESTION PAPER **Total: 50 Marks**